

Culture Quartile Backtest — Weight-Estimation Variants

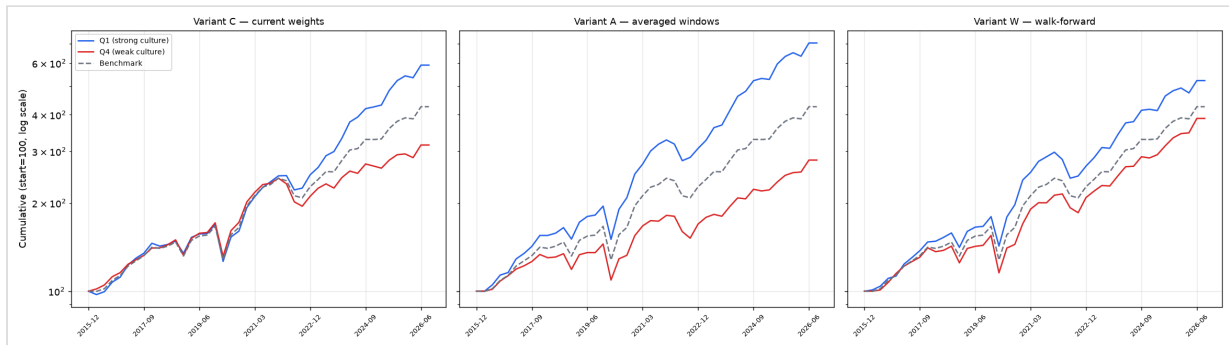
Run date: 2026-07-27 | **Common start:** 2015-12-31 | **Formation quarters:** 43

This study compares three ways of setting the model weights used to rank companies in the quarterly culture-quartile backtest. Everything else (point-in-time review evidence, shrinkage, peer buckets, quartile formation, equal weighting, quarterly rebalancing, prices, benchmark) is identical across variants, so differences are attributable to the weights alone.

- **Variant C — Current:** today's production weights applied at all snapshots (the existing dashboard method).
- **Variant A — Averaged:** a fixed weight vector per peer bucket, the confidence-weighted average of ridge fits from all 12 rolling 5-year windows (2015 through mid-2026).
- **Variant W — Walk-forward:** at each snapshot, weights averaged only over windows completed by that date. Never uses future information — the honest variant.

Bottom line: Variant C's +6.7pp/yr Q1–Q4 spread splits into –1.0pp (2016–2020) vs +13.4pp (2021–present) — concentrated in the years its weights were fitted on. The honest walk-forward variant (W) shows +7.9pp before 2021 but **–0.9pp since 2021**. Much of the current headline spread therefore reflects look-ahead in the weights, not a signal that was usable at the time. The measurement framework — not the fitted weighting — is the defensible claim.

1. Cumulative returns (indexed to 100 at 2015-12-31)



2. Headline statistics

variant	portfolio	cumulative_pct	annualized_pct	avg_companies
C	Q1	490.9	17.97	244.3
C	Q2	342.0	14.83	241.0
C	Q3	283.4	13.32	223.5
C	Q4	215.4	11.28	222.1
C	Benchmark	326.2	14.44	
C	Q1-Q4 spread	275.5	6.69	
A	Q1	603.1	19.89	237.4
A	Q2	334.8	14.65	240.7
A	Q3	267.8	12.88	227.7
A	Q4	180.2	10.06	225.2
A	Benchmark	326.2	14.44	
A	Q1-Q4 spread	422.9	9.83	
W	Q1	423.4	16.64	237.3
W	Q2	316.2	14.18	240.2
W	Q3	281.8	13.27	228.2
W	Q4	288.7	13.46	225.3
W	Benchmark	326.2	14.44	
W	Q1-Q4 spread	134.7	3.18	

3. Split-period comparison (the decisive diagnostic)

Split at 2020-12-31 by formation snapshot. Q1–Q4 spread, annualised:

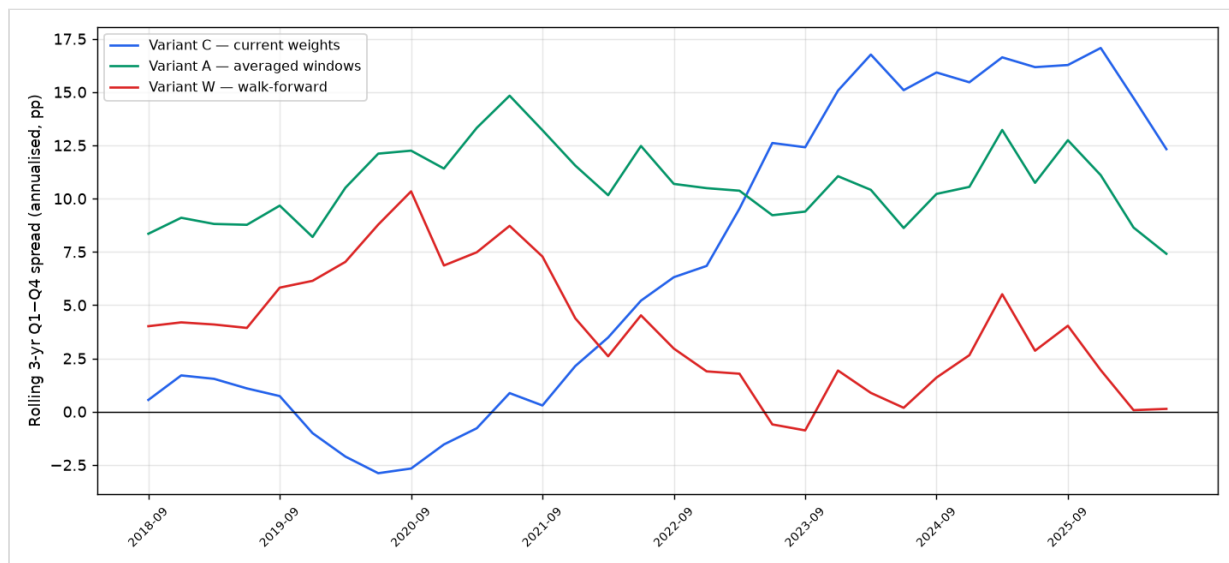
Variant	2016-2020	2021-present	Full period
C — current weights	−1.0pp	+13.4pp	+6.7pp
A — averaged windows	+11.1pp	+8.7pp	+9.8pp
W — walk-forward	+7.9pp	−0.9pp	+3.2pp

Full detail:

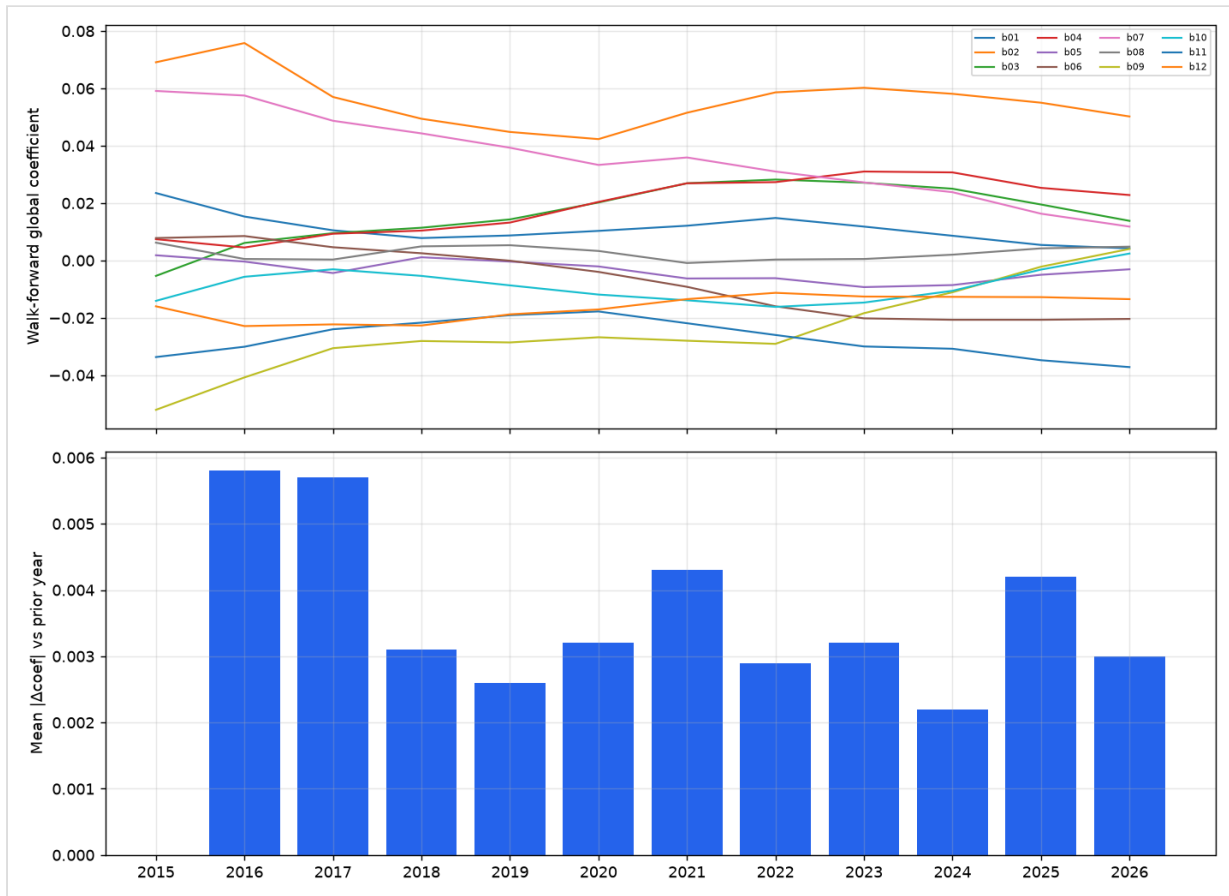
variant	period	portfolio	cumulative_pct	annualized_pct
C	2016-2020	Q1	92.9	14.05
C	2016-2020	Q2	94.5	14.23
C	2016-2020	Q3	92.9	14.05
C	2016-2020	Q4	101.7	15.06
C	2016-2020	Benchmark	95.6	14.35
C	2016-2020	Q1-Q4 spread	-8.8	-1.01
C	2021-present	Q1	206.2	21.49
C	2021-present	Q2	127.2	15.34
C	2021-present	Q3	98.7	12.68
C	2021-present	Q4	56.4	8.09
C	2021-present	Benchmark	117.9	14.51
C	2021-present	Q1-Q4 spread	149.8	13.4
A	2016-2020	Q1	151.0	20.21
A	2016-2020	Q2	100.6	14.94
A	2016-2020	Q3	83.7	12.93
A	2016-2020	Q4	54.7	9.12
A	2016-2020	Benchmark	95.6	14.35
A	2016-2020	Q1-Q4 spread	96.3	11.09
A	2021-present	Q1	180.1	19.62
A	2021-present	Q2	116.8	14.4
A	2021-present	Q3	100.2	12.83
A	2021-present	Q4	81.1	10.88
A	2021-present	Benchmark	117.9	14.51

A	2021-present	Q1-Q4 spread	99.0	8.74
W	2016-2020	Q1	139.5	19.09
W	2016-2020	Q2	91.7	13.9
W	2016-2020	Q3	85.2	13.12
W	2016-2020	Q4	69.8	11.17
W	2016-2020	Benchmark	95.6	14.35
W	2016-2020	Q1-Q4 spread	69.7	7.92
W	2021-present	Q1	118.5	14.56
W	2021-present	Q2	117.1	14.43
W	2021-present	Q3	106.1	13.4
W	2021-present	Q4	129.0	15.5
W	2021-present	Benchmark	117.9	14.51
W	2021-present	Q1-Q4 spread	-10.5	-0.94

4. Rolling 3-year Q1–Q4 spread



5. Walk-forward weight stability (Variant W)



Weights are stable: average year-over-year absolute coefficient change is ~0.003 and shrinking as windows accumulate.

6. Turnover

Variant	Avg quarterly one-way Q1 turnover
C	13.2%
A	12.1%
W	13.6%

Variant W's evolving weights do not cause extra churn.

7. Per-window estimation quality

Global model per window (all 12 windows estimated; no window skipped). Note the steady decline in cross-validated fit as the universe grows —

recent windows carry weaker signal, which is consistent with Variant W's fading post-2021 spread.

Window end	n est.	R ²	CV-R ²
2015-12-31	557	0.247	0.211
2016-12-31	663	0.226	0.195
2017-12-31	751	0.142	0.113
2018-12-31	807	0.148	0.122
2019-12-31	866	0.142	0.115
2020-12-31	894	0.149	0.122
2021-12-31	975	0.226	0.204
2022-12-31	1053	0.157	0.135
2023-12-31	1109	0.147	0.128
2024-12-31	1181	0.097	0.078
2025-12-31	1222	0.066	0.047
2026-06-30	1246	0.056	0.037

8. Caveats

Variant A look-ahead: Variant A is NOT free of look-ahead: every evaluation quarter lies inside at least one estimation window that fed the average. It is a robustness comparison, not the honest benchmark; Variant W is the walk-forward (honest) variant.

- 69 window×bucket cells were flagged thin (estimation set under 60% of today's size), mostly 2015–2017 — early-window results rest on fewer companies.
- Survivorship: universe is today's company list; the same look-ahead/survivorship caveats as the published backtest apply to prices and membership.
- Returns are local-currency, price-only, equal-weighted, no transaction costs.
- Confidence weighting: sum over estimation set of company total scored reviews as of window end. Turnover: one-way: $| \text{names added to Q1} | / | \text{Q1} |$, averaged over rebalances.